

INFORMED CONSENT FOR RESEARCH PARTICIPATION

Neural Organoid Integration Study - Protocol #2026-NO-447

Principal Investigator: Prof. Dr. Gregor Rainer

Institution: University of Fribourg, Department of Medicine, Visual Cognition Laboratory

Study Sponsor: Wyss Center for Bio and Neuroengineering, Geneva

IRB Approval Number: CER-FR-2026-00147

Approval Date: February 15, 2026

PARTICIPANT INFORMATION

Name: _____ Date of Birth: _____

Study ID: _____ Date: _____

Emergency Contact: _____ Phone: _____

PURPOSE OF THIS STUDY

You are being asked to participate in a research study investigating the integration of lab-grown neural organoid tissue for cognitive enhancement. This study aims to demonstrate improved memory consolidation, pattern recognition, and processing speed in healthy adult participants.

Approximately 45 participants will be enrolled across study sites in Fribourg and Geneva.

ELIGIBILITY CRITERIA

To participate in this study, you must meet the following criteria:

Inclusion Criteria:

- Age 21-39 years
- Healthy adult with no history of neurological or psychiatric disorders
- Normal cognitive function (confirmed by baseline testing)
- Fluent in English, French, or German
- Willing to comply with study procedures and follow-up schedule
- Access to reliable internet for remote monitoring
- European residency required for legal and follow-up purposes

Exclusion Criteria:

- History of seizures, brain injury, or neurosurgical procedures
- Current use of psychoactive medications or cognitive enhancers
- Pregnancy, breastfeeding, or planning pregnancy during study period
- Contraindications to MRI (metal implants, claustrophobia)
- Active substance use disorder
- Participation in other interventional research studies
- Immunosuppression or autoimmune conditions that may affect tissue integration

STUDY PROCEDURES

If you agree to participate, you will undergo:

Screening Phase (Week 1-2)

- Medical history review and physical examination
- MRI brain imaging
- Cognitive baseline testing (memory, attention, processing speed)
- Blood work and tissue compatibility screening

Integration Procedure (Week 3)

- Minimally invasive surgical placement of organoid tissue (approximately 10 million neurons, human cortical origin) into the prefrontal or hippocampal region
- The procedure takes approximately 1-2 hours under general anesthesia
- Hospital stay: 2-3 days for monitoring
- Recovery time: 2 weeks before returning to normal activities

Follow-up Phase (Months 1-12)

- Monthly cognitive assessments
- Quarterly MRI scans to monitor integration
- Weekly check-ins via secure app

POTENTIAL BENEFITS

Based on preliminary studies and animal models, participants may experience:

Cognitive Enhancement

- Memory improvement: 40-60% increase in working memory capacity and long-term retention
- Processing speed: 25-35% faster pattern recognition and decision-making
- Learning acceleration: Improved ability to acquire new skills and information
- Emotional regulation: Enhanced stress response modulation and mood stability

Long-term Potential

- Sustained cognitive performance with aging
- Reduced risk of neurodegenerative decline
- Adaptive neural architecture that continues to optimize over time

Please note: These benefits have been observed in preliminary trials but individual results may vary. Some participants may experience minimal or no enhancement.

RISKS AND DISCOMFORTS

Surgical Risks (Standard)

- Infection (1-2% risk)
- Bleeding or blood clots (1-3% risk)
- Adverse reaction to anesthesia (<1% risk)
- Headache, fatigue, or discomfort during recovery

Integration-Specific Risks

Biological Adaptation

I understand that the organoid tissue will:

- ☐ Physically integrate with my existing neural tissue through synaptic formation and vascular innervation
- ☐ Adapt to my neural patterns, neurotransmitter environment, and circadian rhythms
- ☐ Undergo structural changes (growth, pruning, pathway formation) in response to my cognitive activity
- ☐ Form connections that cannot be reversed without potential loss of function

I acknowledge that this adaptation is bilateral - the organoid tissue will influence my cognitive patterns as I influence it.

Measurement Limitations

I acknowledge that:

- ☐ Current neurotechnology can measure neural activity and behavioral outputs but cannot decode subjective experience
- ☐ The organoid tissue may be undergoing internal states (activation patterns, stress responses, or other processes) that cannot be directly observed or interpreted
- ☐ My inability to measure these states does not confirm their absence
- ☐ Enhanced measurement capabilities may be developed during or after my participation

Uncertainty Regarding Agency

I understand that:

- ☐ The organoid tissue will demonstrate learning and adaptation in response to my inputs
- ☐ These adaptive responses may represent: algorithmic optimization, preference formation, distress signaling, or autonomous processing - and current technology cannot distinguish between these possibilities
- ☐ The tissue cannot communicate through language or behavior I can readily interpret
- ☐ Useful cognitive outputs do not confirm the absence of autonomous experience in the tissue
- ☐ I may be unable to determine whether I am using a sophisticated biological tool or engaging with tissue that has some form of agency

Long-term Considerations

I acknowledge:

- ☐ This is a first-in-human study with limited long-term data
- ☐ The effects of bilateral cognitive coupling are not fully understood
- ☐ Integration may result in cognitive changes I cannot predict or control
- ☐ Future advances in consciousness detection technology may provide new information about the nature of organoid tissue that is not available at the time of this procedure

ALTERNATIVES TO PARTICIPATION

Alternative approaches to cognitive enhancement include:

- Pharmacological interventions (nootropics, stimulants)
- Non-invasive brain stimulation (TMS, tDCS)
- Cognitive training programs
- Lifestyle modifications (exercise, sleep optimization, meditation)
- Choosing not to pursue enhancement

VOLUNTARY PARTICIPATION AND WITHDRAWAL

Your participation in this study is completely voluntary. You may choose not to participate or may withdraw at any time without penalty or loss of benefits to which you are otherwise entitled.

Please note: Due to the nature of tissue integration, withdrawal from the study does not guarantee reversal of the procedure. Removal of organoid tissue may result in loss of enhancement effects and potential cognitive disruption.

CONFIDENTIALITY

All information collected will be kept confidential to the extent allowed by law. Your identity will be coded, and only study personnel will have access to identifying information.

Neural activity data and cognitive assessments will be stored in secure, encrypted databases.

COSTS AND COMPENSATION

There is no cost to you for participating in this study. All study-related procedures, including surgery, monitoring, and assessments, are provided at no charge.

You will receive:

- **5.000,00 € upon completion of the integration procedure**
- **100,00 € for each monthly follow-up visit (up to \$1.200,00 € over 12 months)**
- **Coverage of all travel expenses related to study visits**

LIABILITY AND INJURY

In the event of injury resulting from this research, medical treatment is available but will not be provided free of charge. You do not waive any legal rights by signing this consent form.

I release the study investigators and sponsoring institution from liability for:

- ☐ Any experiences occurring in the organoid tissue that cannot be measured or verified with current technology
- ☐ Ethical implications that may emerge as measurement and detection capabilities improve
- ☐ Long-term consequences of integration that cannot be predicted based on available data
- ☐ Psychological or cognitive effects of bilateral coupling with biological tissue

QUESTIONS AND CONTACTS

If you have questions about this study, please contact:

Study Coordinator: Dr. Nina Weber

Phone: +41 26 300 8540

Email: nina.weber@unifr.ch

For questions about your rights as a research participant:

CER-FR (Commission cantonale d'éthique de la recherche sur l'être humain) Office: +41 26 305 7730

CONSENT STATEMENT

I have read this consent form (or it has been read to me). I have had the opportunity to ask questions, and all my questions have been answered to my satisfaction.

I understand:

- The purpose, procedures, risks, and benefits of this study
- That the organoid tissue will adapt to my neural activity through biological processes I cannot directly observe or control
- That current technology cannot determine whether the tissue experiences subjective states
- That I am willing to proceed despite these uncertainties
- That my participation is voluntary and I may withdraw at any time, though reversal of integration is not guaranteed

By signing below, I voluntarily agree to participate in this research study.

Participant Signature: _____ **Date:** _____

Participant Name (printed): _____